Wi-Fi 6 AND Wi-Fi 7 Powering The Next Wave of Smart Connectivity

Wi-Fi 6 leads with faster data rates and reduced latency while upcoming Wi-Fi 7's backward compatibility will facilitate gradual upgrades from Wi-Fi 6, easing transitions. However, advanced features like multi-link operation (MLO) and ultra-low latency may command a premium, making Wi-Fi 7 suited for high-end applications.

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The Internet of Things (IoT) is transforming device communication and driving the demand for faster, more efficient wireless connectivity. Wi-Fi 6 enhances IoT performance in smart homes and industrial automation with higher data rates, improved capacity, and lower latency. Upcoming Wi-Fi 7 will further advance these capabilities, offering greater speeds, reduced congestion, and optimised performance in dense IoT environments. Together, these technologies would revolutionise IoT applications by enabling seamless communication, real-time data processing, and robust connections in complex systems.

IoT fosters digitalisation through data-driven decision-making and supports decarbonisation by reducing energy consumption. It promotes smarter cities and industries, addressing urbanisation and resource challenges. Firmware updates are crucial for IoT device security and performance, with AI and ML minimising user intervention while enhancing efficiency and predictive maintenance. Reliable IoT systems rely on robust hardware and software, with companies like Infineon ensuring quality and security. High-quality hardware, secure updates, and performance optimisation are essential for IoT success.

Digital twins—replicas of physical systems—enable real-time simulation and performance optimisation, allowing engineers to monitor, predict, and enhance system behaviour without physical testing. This approach reduces downtime and boosts efficiency, particularly in manufacturing and logistics. Connectivity is vital for digitalisation and decarbonisation, allowing IoT devices like thermostats, EV chargers, and solar panels to intelligently manage energy consumption, optimising efficiency, and lowering carbon emissions. For instance, smart homes can adjust energy use based on real-time data, contributing to sustainability and energy savings.

Description of Wi-Fi 6, highlighting its key features

Wi-Fi sensing: The next frontier

One of the most exciting developments in Wi-Fi technology is the rise of Wi-Fi sensing. This technology can detect movement and occupancy within a home, opening the door to new levels of